

Scalability & Future Plan

Date	03 July 2026
Team ID	XXXXXX
Project Name	Voice Based concept understanding Analyser
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-threaded local audio file processing.	Simultaneous multi-user uploads cause API latency or backend slowdowns.	High
2	Local SQLite database file storage architecture.	Not suitable for production scaling or distributed system environments.	Medium
3	Basic speech recognition without speaker isolation.	Multiple speakers or heavy background noise degrade transcription accuracy	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles single local user requests.	Deploy on AWS with auto-scaling to manage multiple user requests
2	Data Storage	Local SQLite storage file.	Migrate database layer to cloud-managed PostgreSQL.
3	Performance	Synchronous sequential execution.	Implement Celery with Redis for asynchronous background processing.
4	Security	Hardcoded API keys and no access barriers.	Implement OAuth2 authentication and AWS Secrets Manager.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User Authentication & Dashboard.	1 Month.	Allows instructors to save, manage, and track individual student historical grading records.
Phase 3	Video Analysis & Multi-Language Support.	3 Months	Evaluates non-verbal cues from presentation videos and tracks multilingual speech patterns.
Phase 4	Automated LMS System Integrations.	6 Months	Syncs oral evaluation scores directly to college portals like Canvas or Moodle automatically.